RAG over a Board-Game Rulebook: Bi-Encoder Retrieval vs. Cross-Encoder Re-Ranking

Your name · *Roll number* · *Course* · 19 October 2026

Code: `https://github.com/HimalayanSaswataBose/LMA_A1_2023114004`

Setup and deliverables

Repository	`RAG_Baseline_vs_ReRanking.ipynb` (one commented notebook, saved with all outputs), `results/` (10 CSV/JSON tables and 2 charts written by the notebook), `README.md`, `requirements.txt` (pinned versions).
Document	*Brass: Birmingham* rulebook v2018.11 (Roxley), 12 PDF pages, 2–4 text columns per page; parsed with PyMuPDF into 299 text blocks in reading order (nb. §2).
Corpus	59 chunks cut at sentence boundaries, 47–181 words (median 123); the *same* corpus feeds both pipelines (nb. §3).
Pipeline 1	`sentence-transformers/all-MiniLM-L6-v2` bi-encoder (384-d, L2-normalised) + FAISS `IndexFlatIP` (inner product = cosine) → top-10 → top-3 to the LLM (nb. §4).
Pipeline 2	Same top-10 → `cross-encoder/ms-marco-MiniLM-L-6-v2` scores all 10 (query, chunk) pairs → re-sort by score → top-3 to the LLM (nb. §5).
Generator	`Qwen/Qwen3-4B-Instruct-2507`, local, bf16, greedy decoding, ≤200 new tokens; one shared system prompt and the template "`[Excerpt 1..3] … Question: …`" (nb. §6).
Test queries	5 failure queries F1–F5 covering the four required failure types, 3 limitation queries L1–L3 (nb. §8), and 76 questions labelled with their gold chunk(s) (nb. §9).
Hardware	Intel Core i7-12700F, NVIDIA RTX 5060 Ti 16 GB; Python 3.13, PyTorch 2.8.0 (CUDA 12.9), transformers 5.19.0, sentence-transformers 6.1.0, faiss-cpu 1.15.1. All models run locally.

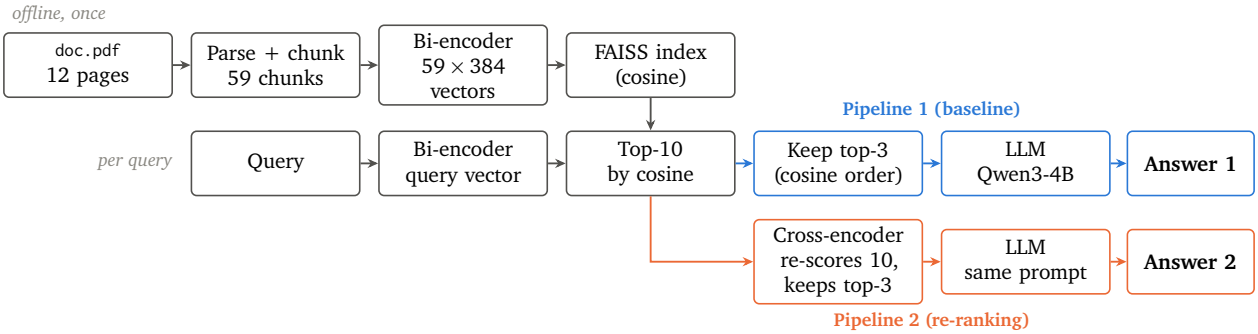


Figure 1: The two pipelines share the chunked corpus, the FAISS index and the first-stage top-10. They differ only in how the three chunks given to the LLM are chosen.

1 Architecture comparison

Writing guide – Section 1. *Asked:* the core difference between a bi-encoder (independent embeddings, cosine similarity) and a cross-encoder (joint self-attention across tokens).

Evidence from this project to draw on:

- Bi-encoder (nb. §4): the 59 chunks were embedded once into a 59 × 384 matrix (0.24 s); each query then needs one encoder pass (4.3 ms GPU) and a 0.02 ms FAISS search. Cosine scores seen in the top-10 lists: 0.17–0.60. Input limit 256 tokens (chunk 1, 311 tokens, is truncated).
- Cross-encoder (nb. §5): nothing is stored; each of the 10 (query, chunk) pairs is one forward pass returning a single relevance logit (seen: −11.4 to +7.4); 9.7 ms for 10 pairs on GPU.
- Both are 6-layer MiniLM models with ≈22.7 M parameters and 384-d hidden states.

2 Qualitative and rank-shift analysis

Table 1: Rank of the gold chunk (the chunk that holds the answer) among the 10 bi-encoder candidates, and the three chunks each pipeline sent to the LLM (gold chunks in bold). Source: `results/rank_shifts.csv`.

	Query	Failure type	Gold	P1 rank	P2 rank	P1 context	P2 context
F1	Can I overbuild an opponent's brewery?	Negation / role limit	43, 53	#10	#1	52, 34, 44	**43**, 34, 52
F2	What happens to level 1 industries at the end of the canal era?	Exception	27	#7	#1	56, 42, 51	**27**, 55, 26
F3	Does my income level give victory points at the end of the game?	Keyword trap	55	#6	#1	24, 57, 30	**55**, 24, 57
F4	What happens when the deck runs out of cards?	Synonyms	21	#5	#1	28, 20, 14	**21**, 28, 20
F5	Is the general supply of cubes limited?	Buried chunk	16	#6	#3	40, 33, 2	32, 43, **16**
L1	Can my opponent use coal from my mine?	Re-ranker demotes	31	#1	#5	**31**, 32, 52	43, 32, 53
L2	What do I get for selling to Warrington with beer?	First-stage miss	46	>10	>10	34, 44, 15	15, 44, 34
L3	Can I upgrade a factory to a better one?	Low CE confidence	42, 43	#1	#2	**42**, 55, 51	30, **43**, 4

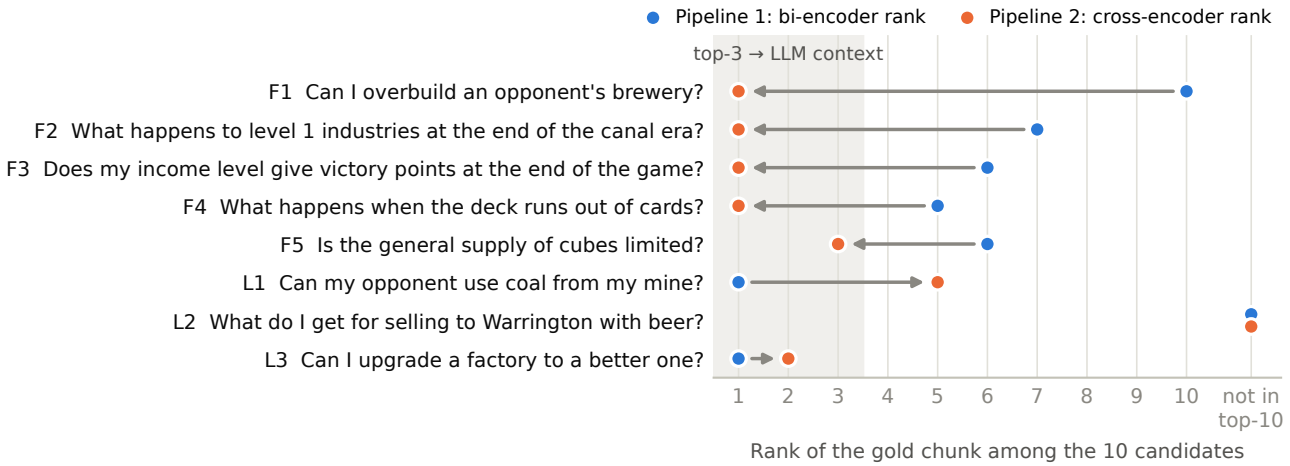


Figure 2: Gold-chunk rank under each pipeline. Arrows run from the bi-encoder rank to the cross-encoder rank; the shaded band is the top-3 passed to the LLM.

Table 2: Answer 1 (baseline) vs. Answer 2 (re-ranked), excerpted verbatim; full answers in `results/answers.csv`. Marks compare each answer with the rulebook: ✓ correct, ◗ partly correct, ✗ wrong, ❍ declined to answer.

	Answer 1 (Pipeline 1)	Answer 2 (Pipeline 2)	Rulebook
F1	❍ "… based on the provided excerpts, this question cannot be answered."	✓ "No … Overbuilding is only allowed for opponent-owned Coal Mines or Iron Works."	No: only an opponent's Coal Mine or Iron Works, and only when no such cubes are left.
F2	✗ "… level 1 industries remain on the board and can be upgraded or replaced during the Rail Era …"	✓ "… all level 1 Industry tiles are removed from the board and returned to the box."	Removed to the box; level 2+ tiles stay.
F3	✓ "No … income level gives victory points only in the introductory game …"	✗ "Yes … you score VPs equal to your income level …"	No; only the introductory game scores income.
F4	◗ "… rounds continue until both the Draw Deck and players' Hands are exhausted. After that, the Rail Era scoring is performed …"	✓ "… players' Hands decrease each round until they have no cards left."	Hands shrink each round; the era ends when they are empty.
F5	✗ "Yes, the general supply of cubes is limited. The rulebook states there are 18 Iron cubes and 30 Coal cubes …"	✓ "No … The General Supply is limitless."	No, the General Supply is limitless.
L1	✓ "Yes … the closest … connected unflipped Coal Mine … (owned by any player)."	✗ "No … coal from a mine belongs to the owner of that mine."	Yes, from any player's connected Coal Mine.
L2	✗ "In a 2-player game, no Merchant tiles are placed in Warrington … you get nothing …"	❍ "… the exact reward … is not specified in the excerpts."	£5 from the Bank.
L3	✓ "Yes … Overbuilding allows you to replace an already placed Industry tile with a higher-level tile …"	❍ "The rulebook does not provide information about upgrading a factory …"	Yes, by Overbuilding.

Table 3: Retrieval quality over the 76 labelled questions (nb. §9). Hit@k: gold chunk within the top k; MRR@10: mean of 1/rank. Right: where the gold chunk landed relative to the LLM's top-3.

	Hit@1	Hit@3	MRR@10	Recall@10
Pipeline 1 (bi-encoder)	0.605	0.816	0.724	0.947
Pipeline 2 (+ cross-encoder)	**0.697**	**0.882**	**0.794**	0.947

Gold chunk vs. top-3	Questions
moved *into* top-3 by re-ranking	8
pushed *out of* top-3 by re-ranking	3
in top-3 under both / neither	59 / 6

Writing guide – Section 2. *Asked:* document the rank changes for the failure queries, and compare Answer 1 vs. Answer 2: did the baseline hallucinate, omit an essential clause or misstate a rule because the critical chunk sat outside the top 3?

Evidence from this project to draw on:

- Table 1 / Fig. 2: every F-case gold chunk starts at rank 5–10 and ends in the top-3; F1 moves furthest (#10 → #1).
- Table 2, F1–F5: Answer 1 declines (F1), states the wrong rule (F2), is partly wrong (F4) and invents cube counts (F5); in F3 it is Answer 2 that is wrong.
- F3: chunk 55 ("income level … is not worth VPs") reaches #1, but chunk 57 (introductory-game scoring) is still in the P2 context.
- L1: chunk 31 (coal from a mine "owned by any player") drops #1 → #5; the cross-encoder ranks chunk 43 (Overbuilding) and 32 (Coal Market) above it.
- L2: chunk 46 (Warrington's £5 bonus) is in neither top-10, so both pipelines see the same three chunks in a different order.
- L3: all ten cross-encoder logits lie between -10.67 and -11.40; in F1–F3 the top logit is $+3.4$ to $+6.9$.
- Table 3: Hit@3 rises 0.816 → 0.882 (8 in, 3 out); Recall@10 is unchanged.

3 Latency profiling

Table 4: Left: retrieval latency per query in ms, median (p95), over 76 questions × 5 repeats (nb. §10). Right: mean end-to-end latency in ms on the GPU for the eight test queries, models already loaded. Sources: `results/latency_retrieval.csv`, `latency_end_to_end.csv`.

Stage	GPU	CPU
Query embedding (bi-encoder)	4.26 (6.57)	8.21 (8.91)
FAISS search, 59 chunks	0.02 (0.03)	0.03 (0.03)
Pipeline 1 retrieval	**4.28** (6.60)	**8.24** (8.94)
Cross-encoder, 10 pairs	9.65 (11.61)	92.03 (115.71)
Pipeline 2 retrieval	**14.24** (17.74)	**100.44** (124.75)
Pipeline 2 / Pipeline 1 (median)	×3.3	×12.2

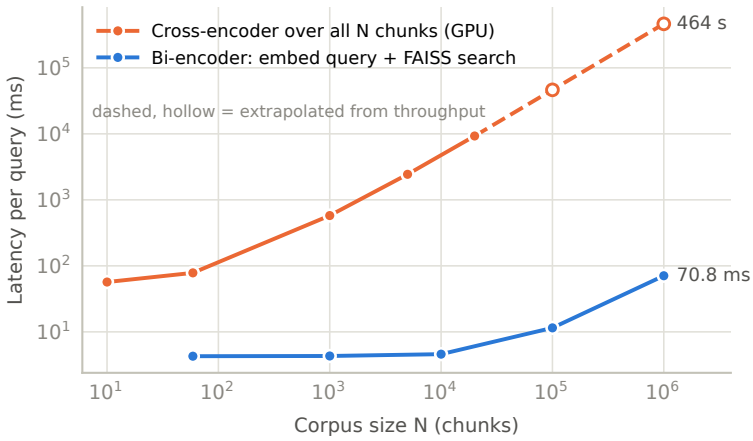
	P1	P2
Retrieval	6.0	6.0
Re-ranking	–	9.9
LLM generation	2288.5	2011.1
Total	**2294.6**	**2027.1**
Answer length (tokens, mean)	52.8	46.0
LLM time per token	42–46 ms	

Writing guide – Section 3. *Asked:* record and tabulate the runtime of the baseline retrieval vs. the re-ranking pipeline (Table 4).

Evidence from this project to draw on:

- Re-ranking adds about 10 ms per query on the GPU and about 92 ms on the CPU.
- Retrieval as a share of end-to-end time: 6.0/2294.6 = 0.26 % (P1) and 15.9/2027.1 = 0.78 % (P2).
- P2's total is lower here while its answers average 46.0 tokens vs. 52.8 for P1, at 42–46 ms per token.

4 Scaling question



N chunks	Bi-enc. + FAISS	Cross-encoder, all N GPU	CPU
59	4.3 ms	78 ms	0.82 s
1 000	4.3 ms	0.58 s	8.5 s*
100 000	11.5 ms	46 s*	14 min*
1 000 000	70.8 ms	7.7 min*	2.4 h*

* extrapolated from the measured throughput: 2 156 pairs/s (GPU), 118 pairs/s (CPU). One-off bi-encoder indexing: 0.55 ms per chunk, ≈9.2 min for 1 M chunks.

Figure 3: Cost of one query as the collection grows (nb. §11). Bi-encoder side: embed the query, then exact FAISS search over N vectors (random unit vectors stand in for $N > 59$). Cross-encoder side: score every (query, chunk) pair, as a cross-encoder-only system would. Sources: `results/scaling_*.csv`.

Writing guide – Section 4. *Asked:* why we do not simply run a cross-encoder across the entire document collection from the very start.

Evidence from this project to draw on:

- Fig. 3: per-query cross-encoder cost grows linearly with N (7.7 min per query on GPU and 2.4 h on CPU at 1 M chunks), against 70.8 ms for query embedding + exact FAISS search.
- The bi-encoder's chunk vectors cost 0.55 ms per chunk, paid once at indexing time; Table 4 shows the per-query part.
- For this 59-chunk rulebook, scoring every chunk takes 78 ms on GPU.
- First-stage Recall@10 = 0.947 (Table 3): the share of questions whose gold chunk reaches the re-ranker at all.